IPCC Tier 2 Cattle GHG Uncertainty Calculator

Technical Methodology: Equations, Monte Carlo Procedure, and Uncertainty Reporting







Version 1.0

Developed under the Global Methane Hub initiative

Contents

1 Introduction **4**
1.1 Purpose . . . 4
1.2 Scope . . . 4
1.3 IPCC Tier Classification . . . 4

2 Input Parameters **5**
2.1 What the defaults assume . . . 5
2.1.1 Which IPCC variant each manure system models . . . 7

3 IPCC Tier 2 Emission Equations **16**
3.1 Net Energy for Maintenance . . . 16
3.2 Net Energy for Activity . . . 16
3.3 Net Energy for Growth . . . 16
3.4 Net Energy for Lactation . . . 17
3.5 Net Energy for Work . . . 17
3.6 Net Energy for Pregnancy . . . 17
3.7 Energy Ratios: REM and REG . . . 17
3.8 Gross Energy Intake . . . 18
3.9 Two Routes to Gross Energy: Modelled and Measured . . . 18
3.9.1 What a measured intake does not replace . . . 19
3.9.2 A residual the method itself carries . . . 19
3.9.3 Reporting which route was used . . . 19
3.10 Enteric Fermentation Emission Factor . . . 19
3.11 Nitrogen Excretion . . . 20
3.12 Volatile Solids and Manure CH_4 . . . 21
3.13 Manure Management N_2O (Direct) . . . 21
3.14 Manure Management N_2O (Indirect) . . . 22
3.15 Pasture/Range/Paddock N_2O (Direct) . . . 22
3.16 Pasture N_2O (Indirect) . . . 23
3.17 GWP-Based CO_2eq Conversion . . . 23

4 Monte Carlo Uncertainty Analysis **24**
4.1 IPCC Approach 2 Rationale . . . 24
4.2 Probability Distributions . . . 24
4.3 Correlated Sampling . . . 25
4.4 Simulation Procedure . . . 26
4.5 Recommended Iteration Count and Convergence . . . 26

5 Uncertainty Metrics **28**
5.1 95% Confidence Interval . . . 28

5.2 Margin of Error (the IPCC reporting metric) . . . 28
5.3 Coefficient of Variation (kept for reference, not reported as headline) . . . 28
5.4 IPCC Table 3.3 Format . . . 28

6 Uncertainty Decomposition **30**

7 Sensitivity Analysis **31**
7.1 Standardised Regression Coefficients (SRC) . . . 31
7.2 Tornado Chart Interpretation . . . 31

8 Correlation Structure **32**
8.1 Activity Data Correlations . . . 32
8.2 Emission Factor Correlations . . . 33
8.3 Structural-Defaults Preset (Literature-Based) . . . 33
8.4 Cross-Block Correlations . . . 35
8.5 Time-Series Correlation Matrix . . . 36
8.6 Across-Group Coefficient Correlation . . . 36
8.7 Animal Numbers Between Groups and Between Years . . . 39
8.8 Year-to-Year Coefficient Correlation (Trend Mode) . . . 40

9 Validation and Quality Assurance **43**

10 Verification of Calculation Accuracy **46**
10.1 Equation-chain verification against manual calculation . . . 46
10.2 Cross-comparison with an independent Monte Carlo package . . . 46
10.3 Automated regression test suite . . . 46
10.4 Verification of IPCC default values and uncertainty ranges . . . 47
10.5 Default uncertainty ranges . . . 47
10.6 Reproducibility . . . 48

11 Reporting **49**
11.1 IPCC Table 3.3 Inputs . . . 49
11.2 UNFCCC Common Reporting Tables Category Map . . . 49
11.3 Disaggregation Levels . . . 50

12 References **51**

1 Introduction

1.1 Purpose

The **IPCC Tier 2 Cattle GHG Uncertainty Calculator** is a web-based tool that enables national inventory compilers to quantify the uncertainty in cattle greenhouse gas (GHG) emission estimates produced using the IPCC Tier 2 methodology. The tool implements the Monte Carlo (Approach 2) uncertainty framework described in IPCC (2006) Volume 1, Chapter 3, applied to the full emission equation chain of IPCC (2006) Volume 4, Chapter 10 and Chapter 11 for pasture N_2O.

Under the Paris Agreement Enhanced Transparency Framework (ETF), national inventory reports are expected to include quantitative uncertainty estimates alongside emission totals. This tool automates that process, from parameter sampling to the formatted IPCC Table 3.3 output, for inventories of GHG emissions from one or more sub-categories of cattle.

1.2 Scope

The tool covers six emission sources for cattle (or buffalo):

1. **Enteric fermentation CH_4**: methane produced during enteric fermentation in the rumen
2. **Manure management CH_4**: methane from anaerobic decomposition of manure in storage systems
3. **Manure management N_2O (direct)**: direct nitrous oxide from nitrogen in managed manure
4. **Manure management N_2O (indirect)**: indirect N_2O via atmospheric deposition and leaching from managed manure
5. **Pasture/range/paddock N_2O (direct)**: direct N_2O from dung and urine deposited on pasture
6. **Pasture/range/paddock N_2O (indirect)**: indirect N_2O from dung and urine on pasture

The tool supports multiple cattle sub-categories (e.g. dairy cows, beef bulls, heifers, growing males, calves) within a single inventory run, with results aggregated to national totals and to per-cattle-type subtotals (dairy, non-dairy), and with a per-sub-category breakdown that users can use to represent production systems, regions, or animal classes.

1.3 IPCC Tier Classification

The tool implements **Tier 2** methods throughout the emission chain. Tier 2 enteric fermentation outputs (GE, DE, CP) feed the manure CH_4 and N_2O calculations, so the whole chain is classified as Tier 2. The uncertainty analysis itself follows the **Approach 2** (Monte Carlo) framework, which is the more accurate of the two IPCC uncertainty methods and is recommended when non-linear equation chains are involved.

2 Input Parameters

All parameters are entered via an Excel template (one row per parameter per cattle sub-category) and loaded into the tool. Each parameter requires a central value, an uncertainty specification (distribution type and bounds or percentage), and an optional IPCC reference.

2.1 What the defaults assume

Every IPCC default this tool ships is one cell of a much larger table. Reaching it means choosing a region, a productivity class, a climate and, for manure, a specific system variant. Those choices are listed below. **Where a choice does not describe the herd being reported, the parameters it governs must be replaced with country-specific values.**

Choice	This tool uses	IPCC also publishes	Affects, and what to do otherwise
Geography	Africa	North America; Western Europe; Eastern Europe; Oceania; Latin America; Asia; Middle East; Indian subcontinent	`BW, MW, WG, Milk, Fat, MilkPR, pct_pregnant, DE, CP, hours`. The tool is built for developing-country inventory compilers. Country-specific values should replace these wherever they exist.
Productivity class	Low productivity	Regional aggregate (Milk 3.5, BW 260); high productivity (Milk 5.8, BW 250)	`BW, Milk, MilkPR, pct_pregnant, DE, CP, Bo, Ym`. Adopted 2026-09-11. Table 10.16A footnote 1 makes low productivity the Tier 1 default for other regions, and the low-productivity row's Pasture/Range feeding situation is the one that matches the activity coefficient the tool uses. Before this the defaults mixed the aggregate and non-dairy rows and described no animal IPCC published.
Feeding situation	Pasture / Range, flat terrain	Stall-fed (Ca 0); grazing large areas or hilly terrain (Ca 0.36)	`Ca`. Matches the low-productivity row, which Annex 10A.1 characterises as Pasture/Range. The tool has no feeding-situation input, so stall-fed or hilly-grazing herds must override the Ca parameter directly.

Choice	This tool uses	IPCC also publishes	Affects, and what to do otherwise
Animal class for manure nitrogen	Other Cattle, for every herd including dairy	Dairy Cow; Swine; Poultry; Other animals	`FRAC`. Table 10.22 publishes a separate Dairy Cow column and this tool does not yet use it, so a dairy herd receives the Other Cattle fractions. The two columns agree for lagoon, liquid slurry, daily spread, dry lot, deep bedding, aerobic treatment and digester. They differ for solid storage (0.45 against 0.30), covered solid storage (0.22 against 0.14) and composting (0.65 against 0.50). A DAIRY HERD USING SOLID STORAGE OR COMPOSTING SHOULD ENTER THE DAIRY COW VALUES FROM TABLE 10.22 rather than accept these.
Climate, soils pathway	Wet	Dry (EF3_PRP 0.002, EF4 0.005, Frac_LEACH_PRP 0); climate-aggregated (EF3_PRP 0.004, EF4 0.010)	`EF3_PRP, EF4, Frac_LEACH_PRP`. DRY-CLIMATE INVENTORIES MUST OVERRIDE THESE THREE. Keeping the wet-climate defaults in a dry climate overstates direct pasture N2O roughly threefold, overstates indirect N2O from deposition almost threefold, and reports a leaching pathway that IPCC treats as absent.
Climate zone, manure methane	Tropical, wherever the tool fills a methane conversion factor for you	Temperate and boreal reference values are on the Vocab sheet; the 2019 Refinement resolves ten zones where this tool resolves four, and its tropical-dry column mirrors tropical	`MCF`. Inventory_Metadata carries no climate field, so any MCF the tool fills in for you assumes a tropical climate. A TEMPERATE OR BOREAL INVENTORY MUST REPLACE EVERY GAP-FILLED MCF: tropical values run roughly two to five times the boreal ones for the same system. Values you enter yourself are used as given.

Choice	This tool uses	IPCC also publishes	Affects, and what to do otherwise
Manure system variant	One IPCC sub-type per manure system, named on every row of the manure-system table	Liquid slurry WITH versus without a natural crust; composting static pile versus in-vessel versus windrow; solid storage plain versus bulking agent versus additives; deep bedding with versus without mixing	`MCF`, `EF3`, `FRAC`. IPCC splits several systems into variants whose coefficients differ by a factor of two or more. Every coefficient on one of our rows comes from the single variant that row declares, so the row describes one real system rather than a blend.
Species	Cattle	Buffalo (Bo 0.10)	`Bo`, `Cp`, `ASH`. The tool is a cattle tool. Buffalo values differ and are not carried.

`MCF`, `EF3` and `FRAC` above are the per-manure-system coefficient families rather than catalogue parameters.

2.1.1 Which IPCC variant each manure system models

IPCC splits several manure systems into variants whose coefficients differ substantially. Liquid slurry with a natural crust emits less methane and volatilises less nitrogen than the same slurry without one, and composting in a static pile is not the same system as composting in a vessel. This tool carries **one row per manure system**, and every coefficient on that row (the methane conversion factor, the direct N_2O emission factor, and the volatilisation and leaching fractions) is read from the single IPCC variant named below, so the row describes a real system rather than a blend. **A herd running a different variant must override the coefficients on that row.**

Manure system	IPCC variant modelled
`pasture`	PRP (Ch.11 pathway; MCF from 2006 Table 10.17 Pasture/Range/Paddock)
`daily_spread`	Daily spread
`solid_storage`	Solid storage (plain)
`solid_storage_covered`	Solid storage - Covered/compacted (2019R Table 10.17)
`dry_lot`	Dry lot
`deep_bedding`	Deep bedding, >1 month accumulation
`liquid_slurry`	Liquid/Slurry, with natural crust cover
`composting`	Composting - Static Pile (forced aeration)

Manure system	IPCC variant modelled
`lagoon`	Uncovered anaerobic lagoon
`anaerobic_digester`	Anaerobic digester, low leakage, high-quality industrial technology, open storage
`aerobic_treatment`	Aerobic treatment, forced aeration
`burned_for_fuel`	Burned for fuel

Table 3 lists all parameters used in the calculation chain, their units, the IPCC equation in which they appear, the default value pre-loaded in the tool (sourced from the IPCC Guidelines), and an indication of potentially suitable probability distributions that can be applied in uncertainty analysis. Where the 2006 Guidelines and the 2019 Refinement give different defaults, both are shown.

Table 3: Input parameters for the IPCC Tier 2 cattle GHG calculation chain: including IPCC defaults and source citations.

Variable	Definition	Unit	IPCC default	IPCC reference (equation / table)	Typical dist.
`N`	Number of animals (population)	head	*n/a*	Eq. 10.1 (Tier 1 totals); national activity data	Normal
`BW`	Mean live body weight	kg/head	270	Eq. 10.3; Annex 10A.1 (Africa, low productivity)	Normal
`Cfi`	Coefficient for NE_m	MJ/kg$^{0.75}$/d	0.386 (lact.) / 0.322 (non-lact.) / 0.370 (bulls)	Eq. 10.3; Table 10.4 (Updated)	Constant
`Tw`	Mean winter temperature	°C	20 (inert)	Eq. 10.2 (cold-climate Cfi adjustment)	Constant
`Ca`	Activity coefficient	:	0.00 (stall) / 0.17 (pasture flat) / 0.36 (large/hilly)	Eq. 10.4; Table 10.5 (Updated)	Constant
`WG`	Mean daily weight gain	kg/head/d	0 (non-growing default)	Eq. 10.6	Normal
`MW`	Mature body weight	kg/head	300	Eq. 10.6; Annex 10A.2	Normal

Continued on next page

Table 3 continued

Variable	Definition	Unit	IPCC default	IPCC reference (equation / table)	Typical dist.
`C`	Sex/condition coefficient for NE_g	:	0.8 (female) / 1.0 (castrate) / 1.2 (bull)	Eq. 10.6	Constant
`Milk`	Daily milk yield	kg/head/d	1.2 (Africa, low productivity)	Eq. 10.8; Annex 10A.1 (Africa, low productivity)	Normal/Lognormal
`MilkFat`	Milk fat content	%	4.3 (Africa, IPCC 2019R)	Eq. 10.8; Annex 10A.1	Normal
`pct_pregnant`	Fraction of females pregnant / yr	:	0.52	Annex 10A.1 (Africa, low productivity)	Beta
`MilkPR`	Milk protein content	%	3.6 ($\approx 1.9 + 0.4\times$ %Fat at Fat 4.3)	Eq. 10.33 (uses 6.38 milk-protein-to-N); Annex 10A.1–10A.3	Normal
`WorkHours`	Daily working hours	h/head/d	0 (no work)	Eq. 10.11	Constant
`Cp`	Pregnancy coefficient	:	0.10 (cattle)	Eq. 10.13; Table 10.7 (Updated)	Constant

Continued on next page

Table 3 continued

Variable	Definition	Unit	IPCC default	IPCC reference (equation / table)	Typical dist.
DE_pct	Digestible energy (% of GE)	%	51.0 (Africa, low productivity)	Eq. 10.14–16; Annex 10A.1 (Africa, low productivity)	Normal
Ym	Methane conversion factor	%	6.5 (lactating dairy cows, the catalogue default); resolved per subcategory: 7.0 non-dairy and multi-purpose, 4.0 feedlot under 2019R; 6.5 and 3.0 respectively under 2006	Eq. 10.21; Table 10.12 (Updated), footnote 4	Normal/PERT
UE	Urinary energy (fraction of GE)	:	0.04 (cattle)	Eq. 10.24 (footnote)	Constant

Continued on next page

Table 3 continued

Variable	Definition	Unit	IPCC default	IPCC reference (equation / table)	Typical dist.
`ASH`	Ash content of manure (fraction of DMI)	:	0.08 (cattle; IPCC example 0.06 is pig-specific)	Eq. 10.24	Constant
`MMS_fractions`	Fraction of manure in each MMS (deterministic; constrained to sum to 1 per sub-category). Optional per-row `lower_fraction` / `upper_fraction` / `distribution_fraction` columns on the Manure_Management sheet enable per-iteration sampling within the user's bounds, renormalised to preserve the simplex; sampled fractions surface as `fraction_<mms>` in the sensitivity tornado.	:	country-specific (Worksheet 10.A-9)	Eq. 10.23	Deterministic (default) or per-MMS PERT/Normal/etc. with renormalisation

Continued on next page

Table 3 continued

Variable	Definition	Unit	IPCC default	IPCC reference (equation / table)	Typical dist.
`Bo`	Maximum CH_4 producing capacity	m^3 CH_4/kg VS	0.13 (2019R "Other regions, low productivity" cattle); 2006 Africa = 0.10	Eq. 10.23; Table 10.16(a) (Updated)	Normal
`MCF`	Methane conversion factor (MMS)	:	per-MMS & climate (e.g. 80% lagoon tropical)	Eq. 10.23; Table 10.17 (Updated)	Normal/PERT
`EF3`	Direct-N_2O EF (per MMS)	kg N_2O-N/kg N	per-MMS (e.g. 0.010 solid storage)	Eq. 10.25; Table 10.21 (Updated)	Lognormal/PERT
`Frac_GasMS`	Fraction N volatilised (MMS)	:	per-MMS, e.g. 0.45 solid storage (2019R)	Eq. 10.26; Table 10.22 (Updated)	PERT (preferred, IPCC ranges are asymmetric) / Normal

Continued on next page

Table 3 continued

Variable	Definition	Unit	IPCC default	IPCC reference (equation / table)	Typical dist.
`EF4`	EF for atmospheric deposition	kg N_2O-N/(kg NH_3-N+NO_x-N)	**0.014** (2019R wet climate, the shipped default); aggregated 0.010, dry 0.005; 2006 = 0.010	Eq. 10.27; Vol.4 Ch.11 Table 11.3 (Updated)	PERT
`Frac_LeachMS`	Fraction N leached (MMS)	:	per-MMS, e.g. 0.02 solid storage	Eq. 10.28; Table 10.22 (Updated)	PERT
`EF5`	EF for leaching/runoff	kg N_2O-N/kg N	0.011 (2019R); 2006 = 0.0075	Eq. 10.29; Vol.4 Ch.11 Table 11.3 (Updated)	PERT
`Frac_PRP`	Fraction of dung and urine deposited on pasture	:	country-specific	Vol.4 Ch.11 Eq. 11.5 ($MS_{(T,PRP)}$)	Beta / PERT (assignable via the `distribution` column)

Continued on next page

Table 3 continued

Variable	Definition	Unit	IPCC default	IPCC reference (equation / table)	Typical dist.
`EF3_PRP`	Direct N_2O EF, pasture (cattle/poultry/pigs)	kg N_2O-N/kg N	**0.006** (2019R wet climate, the shipped default); aggregated 0.004, dry 0.002; 2006 = 0.02	Vol.4 Ch.11 Eq. 11.1; Table 11.1 (Updated)	PERT
`Frac_GasPRP`	Fraction N volatilised (PRP)	:	0.21 (2019R FracGASM); 2006 = 0.20	Vol.4 Ch.11 Eq. 11.9; Table 11.3 (Updated)	PERT (preferred, IPCC ranges are asymmetric) / Normal
`Frac_LeachPRP`	Fraction N leached (PRP)	:	0.24 (2019R wet); 2006 = 0.30; dry = 0	Vol.4 Ch.11 Eq. 11.10; Table 11.3 (Updated)	PERT

3 IPCC Tier 2 Emission Equations

This section presents the complete equation chain implemented in the tool, following IPCC (2006) Volume 4, Chapter 10. All equations are evaluated for each cattle sub-category independently; results are then aggregated to national totals by summing across sub-categories.

3.1 Net Energy for Maintenance

The daily net energy requirement for maintenance (NE_m, MJ head^{-1} day^{-1}) is given by IPCC Eq. 10.3:

$$NE_m = C_{fi,\text{adj}} \times BW^{0.75}$$

where BW is the mean live body weight (kg head^{-1}) and C_{fi} is a sex/condition coefficient (MJ kg$^{-0.75}$ day^{-1}). For open-lot cattle in colder climates the 2006 Guidelines (Vol.4 Ch.10, retained unchanged in the 2019 Refinement) prescribe a linear adjustment of C_{fi}: IPCC Eq.~10.2, "Coefficient for calculating net energy for maintenance'', based on the mean daily winter temperature:

$$C_{fi,\text{adj}} = C_{fi} + 0.0048 \times (20 - T_w) \quad \text{when } T_w < 20°\text{C}$$

otherwise $C_{fi,\text{adj}} = C_{fi}$. This adjustment increases the maintenance energy requirement by approximately 0.48% per degree below 20°C. The same linear formula is implemented in the IPCC Inventory Software (v2.95).

3.2 Net Energy for Activity

The net energy for physical activity (NE_a, MJ head^{-1} day^{-1}) is given by IPCC Eq. 10.4:

$$NE_a = C_a \times NE_m$$

where C_a is a dimensionless activity coefficient that accounts for the terrain and movement patterns of the animal (e.g. 0.00 for housed animals, 0.17 for animals on flat pasture, 0.36 for large areas or hilly terrain).

3.3 Net Energy for Growth

The net energy required for growth (NE_g, MJ head^{-1} day^{-1}) is given by IPCC Eq. 10.6:

$$NE_g = 22.02 \times \left(\frac{BW}{C \times MW}\right)^{0.75} \times WG^{1.097}$$

where WG is the mean daily weight gain (kg head^{-1} day^{-1}), MW is the mature body weight (kg),

and C is a sex/physiological condition coefficient (0.8 for females, 1.0 for castrates, 1.2 for bulls). $NE_g = 0$ when $WG \leq 0$.

3.4 Net Energy for Lactation

The daily net energy for milk production (NE_l, MJ head^{-1} day^{-1}) is given by IPCC Eq. 10.8:

$$NE_l = \text{Milk} \times (1.47 + 0.40 \times \text{MilkFat})$$

where Milk is the daily milk yield (kg head^{-1} day^{-1}) and MilkFat is the fat content (%). For non-lactating sub-categories, set Milk = 0.

3.5 Net Energy for Work

The net energy for work (NE_w, MJ head^{-1} day^{-1}) is given by IPCC Eq. 10.11 (used when cattle perform physical work, e.g. traction or load-carrying):

$$NE_w = 0.10 \times NE_m \times \text{WorkHours}$$

where WorkHours is the mean number of hours of work per day. Set to 0 when animals do not perform work.

3.6 Net Energy for Pregnancy

The net energy allocated to fetal growth during pregnancy (NE_p, MJ head^{-1} day^{-1}) is given by IPCC Eq. 10.13:

$$NE_p = C_p \times NE_m$$

where C_p is a coefficient for the energy cost of pregnancy (0.10 for cattle). The parameter `pct_pregnant` can pro-rate C_p when not all females in a sub-category are pregnant in the same year ($0 \leq$ pct_pregnant ≤ 1). Includes pregnant heifers that have not yet calved. The earlier name `pct_calving` is accepted as an alias on upload.

3.7 Energy Ratios: REM and REG

The ratio of net energy for maintenance to digestible energy consumed (REM) is given by IPCC Eq. 10.14:

$$REM = 1.123 - (4.092 \times 10^{-3} \times DE) + (1.126 \times 10^{-5} \times DE^2) - \frac{25.4}{DE}$$

The ratio of net energy for growth to digestible energy consumed (REG) is given by IPCC Eq. 10.15:

$$REG = 1.164 - (5.160 \times 10^{-3} \times DE) + (1.308 \times 10^{-5} \times DE^2) - \frac{37.4}{DE}$$

where DE is the digestible energy expressed as a percentage of gross energy. Both equations are empirical fits derived from ruminant data and are only valid over the digestibility range observed for ruminant diets. The 2019 Refinement (Vol.4 Ch.10, §10.2) documents that range as **45–85%** (45–55% for crop by-products and rangelands; 55–80% for managed pasture and forage-based diets; 72–85% for grain-based feedlot diets). Outside this range the fits extrapolate poorly: REG passes through zero near $DE \approx 37.6\%$, and below that the $-37.4/DE$ term dominates and drives REG negative. Because the gross-energy equation (below) divides by REM and REG, an unconstrained low-DE draw makes gross energy, and hence emissions, diverge or even turn negative.

To keep every Monte Carlo iteration inside the equations' valid domain, sampled DE values are **clamped to** $[45, 85]\%$ at the sampling stage (`generate_mc_samples`). This prevents non-physical iterations (the extreme positive and negative emission totals that would otherwise stretch the output histogram and pull the simulated mean above the 97.5^{th} percentile) without materially shifting the central estimate, since the clamp only trims the small fraction of the DE distribution that falls outside the documented ruminant range. The earlier $DE > 0$ check is retained as defence-in-depth.

3.8 Gross Energy Intake

The mean daily gross energy intake (GE, MJ head^{-1} day^{-1}) is given by IPCC Eq. 10.16:

$$GE = \frac{(NE_m + NE_a + NE_l + NE_w + NE_p)/REM + NE_g/REG}{DE/100}$$

3.9 Two Routes to Gross Energy: Modelled and Measured

Equation 10.16 above infers intake from what the animal does. Its inputs are animal performance: body weight, growth, milk yield, pregnancy and activity. This is the route the tool takes by default, and the only route earlier versions offered.

Many datasets work in the opposite direction. They measure feed intake and feed quality, and so already hold GE, or hold dry matter intake (DMI) and can convert. Such datasets frequently carry no milk yield or weight gain at all, because nothing in their method required inferring intake. Measured intake is not a lesser input than modelled intake; where it exists it is generally the stronger evidence.

The tool therefore accepts GE or DMI as optional inputs, **per sub-category**:

1. If a usable GE is supplied for a sub-category, it is used directly and Equations 10.3 to 10.16 are not evaluated for that sub-category.
2. Otherwise, if a usable DMI is supplied, $GE = DMI \times 18.45$.
3. Otherwise GE is derived from Equation 10.16 exactly as above.

A value is "usable" only if it is present, finite and strictly positive. Blank, missing, zero and negative all mean "not supplied", so a zero intake is never mistaken for a measurement. Because the choice is made per sub-category, one inventory may mix measured and modelled groups, and an inventory that supplies neither is numerically unchanged from earlier versions of the tool.

GE takes precedence over DMI. A supplied GE carries the dataset's own measured dietary energy density, whereas converting DMI imposes the IPCC figure of 18.45 MJ kg^{-1} dry matter. Where both are given, using GE preserves the user's measurement.

3.9.1 What a measured intake does not replace

Supplying GE bypasses the energy balance only. Three groups of inputs remain necessary, and this is a frequent source of confusion:

- **Digestible energy** DE is still required. With GE supplied it no longer affects enteric methane, but Equation 10.24 still uses it to set the indigestible share of the feed that becomes volatile solids, and hence manure methane.
- **Milk yield, milk protein content and weight gain** are still required. Equations 10.32 and 10.33 obtain nitrogen excretion by subtracting the nitrogen retained in milk and in liveweight gain from the nitrogen eaten. These inputs therefore continue to determine direct and indirect N_2O from both managed manure and pasture.
- **Crude protein** CP is still required, for the same nitrogen-intake term.

3.9.2 A residual the method itself carries

Equations 10.24 and 10.32 divide by the IPCC constant of 18.45 MJ kg^{-1} dry matter regardless of the user's own feed. A dataset whose measured energy density differs, for instance the 16.5 to 17.6 MJ kg^{-1} typical of tropical forage-based systems, will therefore see that constant applied in the volatile-solids and nitrogen-intake terms even while its own GE is used verbatim for enteric methane. This is a property of the IPCC equations, not a defect in the implementation, and it is documented here so it is not mistaken for one.

3.9.3 Reporting which route was used

Every run records, for each sub-category, whether gross energy was modelled or measured. The label appears in the results table in the application, in the run-settings sheet of the Excel export, and in the Word report. Where an inventory mixes routes, the split is stated explicitly rather than a single method being implied.

3.10 Enteric Fermentation Emission Factor

The per-head per-day methane emission factor for enteric fermentation (EF_1, kg CH_4 head^{-1} day^{-1}) is given by IPCC Eq. 10.21:

$$EF_1 = \frac{GE \times (Y_m/100)}{55.65}$$

where Y_m is the methane conversion factor expressed as a percentage of gross energy converted to methane (the /100 converts Y_m to a fraction) and 55.65 MJ kg^{-1} is the energy content of methane. The annual per-sub-category enteric emission is:

$$E_{\text{enteric}} = EF_1 \times N \times 365/1000 \quad (\text{tonnes CH}_4\ \text{yr}^{-1})$$

3.11 Nitrogen Excretion

The daily nitrogen excretion rate (N_{ex}, kg N head^{-1} day^{-1}) follows IPCC Eq. 10.31:

$$N_{\text{ex}} = N_{\text{intake}} - N_{\text{retained}}$$

Nitrogen intake from feed is:

$$N_{\text{intake}} = \frac{GE}{18.45} \times \frac{\text{CP\%}}{6.25 \times 100}$$

where CP% is the crude-protein content of the diet (the template column is labelled `CP`; the value supplied is the same percentage value used here).

Nitrogen retained in body tissue and milk are calculated separately. For milk, the nitrogen content uses the **Jones factor for casein (6.38)**: distinct from the generic protein factor (6.25) used for feed and body tissue. The 6.38 factor and the 6.25 factor both appear inside IPCC 2006 Vol.4 Ch.10 Eq. 10.33 (N retention rates for cattle); the same equation is retained, with updated form, in the 2019 Refinement (also numbered 10.33):

$$N_{\text{milk}} = \frac{\text{Milk} \times \text{MilkProtein\%}}{6.38 \times 100}$$

For growing animals, IPCC Eq. 10.33 adds a body-tissue retention term:

$$N_{\text{growth}} = \frac{WG \times (268 - 7.03 \times NE_g/WG)}{1000 \times 6.25}$$

so that $N_{\text{retained}} = N_{\text{milk}} + N_{\text{growth}}$. For non-growing sub-categories ($WG \leq 0$) the growth term is set to zero.

The annual nitrogen excreted per sub-category ($N_{\text{ex,annual}}$, kg N yr^{-1}) is:

$$N_{\text{ex,annual}} = N_{\text{ex}} \times N \times 365$$

3.12 Volatile Solids and Manure CH_4

Total manure CH_4 emissions are given by IPCC 2006 Vol.4 Ch.10 Eq. 10.22 (the top-level emissions equation, EF × population × 10^{-6}). The per-head per-year emission factor is given by Eq. 10.23:

$$EF_{\mathrm{mm,CH_4}} = VS \times 365 \times B_o \times 0.67 \times \sum_s (MS_s \times MCF_s) \quad \text{(Eq. 10.23)}$$

The daily volatile-solid excretion rate VS (kg dry-matter head^{-1} day^{-1}) that feeds Eq. 10.23 is given by IPCC Eq. 10.24:

$$VS = [GE \times (1 - DE/100) + (UE \times GE)] \times \left(\frac{1 - \mathrm{ASH}}{18.45}\right) \quad \text{(Eq. 10.24)}$$

where UE is the fraction of GE excreted as urinary energy (default 0.04 for cattle, IPCC Eq. 10.24 footnote) and ASH is the ash content of manure (default 0.08 for cattle).

where B_o is the maximum methane producing capacity (m^3 CH_4 kg^{-1} VS), 0.67 is the conversion factor from m^3 CH_4 to kg CH_4 (the density of methane at standard conditions, kg m^{-3}), MS_s is the fraction of manure managed in system s, and MCF_s is the methane conversion factor for system s (fraction of B_o). The 2019 Refinement Table 10.17 (Updated) provides MCF defaults across 11 manure management systems: eight inherited from the 2006 Guidelines plus four added in 2019 (anaerobic digester / biogas, aerobic treatment, solid storage – covered/compacted, and burned for fuel). Default B_o values for cattle are given in Table 10.16a. Per-MMS uncertainty in MCF and in the manure fractions MS_s is sampled independently, when the user supplies optional `lower_fraction` / `upper_fraction` / `distribution_fraction` columns on the Manure_Management sheet, each Monte Carlo iteration's MS_s row is renormalised after sampling so the simplex constraint $\sum_s MS_s = 1$ is preserved exactly.

3.13 Manure Management N_2O (Direct)

Direct N_2O emissions from manure management are given by IPCC 2006 Vol.4 Ch.10 Eq. 10.25 (the indirect-pathway losses are calculated separately in Eq. 10.26–10.29; see next sub-section):

$$E_{\mathrm{direct,mm}} = N_{\mathrm{ex,annual}} \times EF_3 \times \frac{44}{28} \quad (\text{kg } N_2O \text{ yr}^{-1})$$

where EF_3 is the emission factor for direct N_2O (kg N_2O-N kg^{-1} N) and 44/28 converts N_2O-N to N_2O. For multiple MMS systems, EF_3 is computed as the weighted average of per-MMS factors:

$$EF_3 = \sum_s MS_s \times EF_{3,s}$$

3.14 Manure Management N_2O (Indirect)

Indirect N_2O from managed manure arises from two pathways. Each pathway is split between an N-loss equation (in Vol.4 Ch.10) and an N_2O equation (in Vol.4 Ch.10, drawing EFs from Vol.4 Ch.11 Table 11.3):

Atmospheric deposition of volatilised N: Frac_GasMS from Eq. 10.26 multiplied by EF_4 in Eq. 10.27:

$$E_{\text{vol,mm}} = N_{\text{ex,annual}} \times \text{Frac_GasMS} \times EF_4 \times \frac{44}{28}$$

Leaching and runoff: Frac_LeachMS from Eq. 10.28 multiplied by EF_5 in Eq. 10.29:

$$E_{\text{leach,mm}} = N_{\text{ex,annual}} \times \text{Frac_LeachMS} \times EF_5 \times \frac{44}{28}$$

where EF_4 (kg N_2O-N kg^{-1} NH_3-N + NO_x-N volatilised) and EF_5 (kg N_2O-N kg^{-1} N leached/runoff) are from IPCC Vol.4 Ch.11 Table 11.3. In the 2006 Guidelines $EF_4 = 0.010$ (range 0.002–0.05) and $EF_5 = 0.0075$ (range 0.0005–0.025). The 2019 Refinement keeps the aggregated $EF_4 = 0.010$ (range 0.002–0.018) and additionally provides climate-disaggregated values of 0.014 in wet climates (range 0.011–0.017) and 0.005 in dry climates (range 0.000–0.011); EF_5 is updated to 0.011 (range 0.000–0.020) with no climate disaggregation. Frac_GasMS and Frac_LeachMS are MMS- and country- or climate-specific fractions (defaults in Vol.4 Ch.10 Tables 10.22 and 10.23 of the 2019 Refinement).

3.15 Pasture/Range/Paddock N_2O (Direct)

The nitrogen deposited on pasture by grazing animals (F_{PRP}) is the per-sub-category form of IPCC Vol.4 Ch.11 Eq.~11.5 (N in urine and dung deposited by grazing animals on PRP):

$$F_{\text{PRP}} = N_{\text{ex,annual}} \times \text{Frac_PRP}$$

where Frac_PRP is the fraction of total annual N excretion deposited on pasture (denoted $MS_{(T,\text{PRP})}$ in IPCC notation). Direct N_2O from pasture/range/paddock falls under managed soils (IPCC Volume 4, Chapter 11), not Chapter 10. It is given by IPCC Eq.~11.1 (with F_{PRP} as the input N source):

$$E_{\text{direct,PRP}} = F_{\text{PRP}} \times EF_{3,\text{PRP}} \times \frac{44}{28}$$

where $EF_{3,\text{PRP}}$ is the emission factor for direct N_2O from PRP given in IPCC Vol.4 Chapter 11, Table 11.1. The 2006 value is 0.02 kg N_2O-N kg^{-1} N (cattle, poultry, pigs combined). The 2019 Refinement updates this to $EF_{3PRP,CPP}$ for cattle, poultry and pigs with an aggregated default of 0.004 kg N_2O-N kg^{-1} N (range 0.000–0.014). When the climate is known, the disaggregated values are 0.006 in wet climates (range 0.000–0.027) and 0.002 in dry climates (range 0.000–0.007).

3.16 Pasture N_2O (Indirect)

Indirect N_2O from pasture deposition follows IPCC Vol.4 Ch.11 Eq. 11.9 (volatilisation pathway) and Eq. 11.10 (leaching/runoff pathway), using the same pathway structure as managed manure:

$$E_{\text{indirect,PRP}} = F_{\text{PRP}} \times (\text{Frac_GasPRP} \times EF_4 + \text{Frac_LeachPRP} \times EF_5) \times \frac{44}{28}$$

On the soil side of Eq.~11.9/11.10 the same role played by Frac_GasMS and Frac_LeachMS for managed manure is played by FracGASM and FracLEACH-(H) in IPCC Vol.4 Ch.11 Table 11.3: Frac_GasPRP corresponds to FracGASM (the fraction of applied/deposited N volatilised as NH_3 and NO_x) and Frac_LeachPRP corresponds to FracLEACH-(H) (the fraction leached/runoff in regions with annual precipitation exceeding evapotranspiration).

Per IPCC Vol.4 Ch.11, where evapotranspiration exceeds precipitation, leaching may be assumed not to occur: in the tool, `Frac_LeachPRP` may be set equal to zero in those climate cases.

3.17 GWP-Based CO_2eq Conversion

All emission pathways are converted to CO_2 equivalents using Global Warming Potentials (GWP). The tool supports three IPCC assessment report baselines:

Basis	**CH_4 GWP**	**N_2O GWP**	**Reference**
AR4 (100-yr)	25	298	IPCC AR4 WG1, Ch.2 Table 2.14
AR5 (100-yr)	28	265	IPCC AR5 WG1, Ch.8 Table 8.7
AR6 (100-yr, non-fossil CH_4)	27.0	273	IPCC AR6 WG1, Ch.7 Table 7.15 (and Table 7.SM

Total CO_2eq = (total CH_4 tonnes $\times$ GWP_{CH_4}) + (total N_2O tonnes $\times$ GWP_{N_2O}).

4 Monte Carlo Uncertainty Analysis

4.1 IPCC Approach 2 Rationale

The IPCC (2006) Volume 1, Chapter 3 defines two approaches to quantifying inventory uncertainty:

- **Approach 1 (error propagation):** Combines individual parameter uncertainties using simple algebraic rules. Suitable when the emission equation is a product or sum of independent terms. Underestimates uncertainty when equations are non-linear or inputs are correlated.
- **Approach 2 (Monte Carlo simulation):** Propagates uncertainty through the full equation chain by repeatedly sampling all parameters from their distributions. Exact for any equation structure. Recommended for Tier 2 livestock inventories because the GE equation chain is non-linear in BW, WG, and $DE_{\%}$.

This tool implements Approach 2. The simulation samples all input parameters simultaneously for each iteration, evaluates the full equation chain from NE_m through to total CO_2eq, and records the output. The resulting distribution of outputs characterises the uncertainty.

4.2 Probability Distributions

Each parameter is assigned a probability distribution reflecting the state of knowledge about its true value. The tool supports:

Distribution	Parameterisation	Typical use
Normal	mean μ, SD σ	Body weight, milk yield, population counts with symmetric survey error
Positive-Normal	Normal truncated at zero	Non-negative symmetric parameters where negative draws are implausible
Lognormal	log-mean, log-SD (fitted from mean and CV%)	Strictly positive, right-skewed; milk yield in heterogeneous herds; some EFs
Beta	rescaled Beta(α, β) matching the supplied mean and bounds	Fractions in $[0, 1]$ with mode strictly in the interior; pct_pregnant, Cp
Truncated-Normal [0,1]	Normal clipped to $[0, 1]$	Fractions in $[0, 1]$ when a symmetric error around the mean is appropriate
PERT	minimum, most-likely, maximum	Emission factors with expert-elicited bounds; EF3_PRP, EF4, EF5, Bo, MCF
Triangular	minimum, most-likely, maximum	Similar to PERT but with sharper tails; Ca, C_growth
Uniform	lower, upper	Parameters with only the plausible range known; MMS fractions
Constant	fixed value	Parameters treated as known exactly (uncertainty_pct = 0)

When the user supplies `uncertainty_pct` (a symmetric 95% half-width as a percentage of the mean), the tool derives the distribution parameters as:

$$\sigma = \frac{\text{uncertainty_pct} \times \mu}{100 \times 1.96}$$

for a normal distribution, or equivalent transformations for lognormal and PERT.

4.3 Correlated Sampling

When parameters are correlated (e.g. body weight and mature weight are positively correlated; diet quality and methane conversion factor may be negatively correlated), ignoring correlation underestimates or overestimates combined uncertainty. The tool applies the same correlated-sampling procedure to every path that supplies a correlation matrix: the structural-defaults preset, a manually uploaded matrix, and the matrix estimated automatically from a time-series upload all flow through the same algorithm.

Procedure (per IPCC Volume 1 Chapter 3 §3.2.3.2):

1. For each parameter j, draw n_{iter} independent samples from the parameter's own marginal distribution (normal, PERT, lognormal, beta, triangular, uniform, or constant, exactly as specified in the template).
2. Reorder the columns of the resulting $n_{\text{iter}} \times p$ sample matrix so that the realised Spearman rank correlation between columns matches the target matrix $\mathbf{\Sigma}$ (implemented via `mc2d::cornode`, which performs the restricted-pairing operation IPCC §3.2.3.2 cites under "restricted pairing methods that are included in many software packages'').

This procedure is distribution-free: the marginal shapes are preserved exactly, regardless of whether the user has mixed normal / lognormal / PERT / beta parameters in the same correlation block. The target $\mathbf{\Sigma}$ is interpreted as a Spearman rank correlation matrix throughout, the same convention the time-series matrix is computed under (Spearman rank correlation of the user's annual values, after detrending; see Section "Correlation Structure'').

If $\mathbf{\Sigma}$ is not positive definite (e.g. a manually entered matrix that turns out to be inconsistent), the tool projects it onto the nearest positive-definite correlation matrix via `Matrix::nearPD(corr = TRUE)` and emits a warning whenever any individual entry is shifted by more than 0.01. The tool does *not* silently fall back to uncorrelated sampling.

The tool supports both within-block and across-block correlations. The default scope is independence; a single unified correlation matrix spanning AD + coefficients is used whenever cross-block pairs are specified.

4.4 Simulation Procedure

For each Monte Carlo iteration $i = 1, \ldots, n_{\text{iter}}$:

1. Sample all parameters from their distributions (with correlation structure if supplied).
2. For each cattle sub-category k:
   a. Compute $NE_m^{(k)}$, $NE_a^{(k)}$, $NE_g^{(k)}$, $NE_l^{(k)}$, $NE_w^{(k)}$, $NE_p^{(k)}$.
   b. Compute $REM^{(k)}$ and $REG^{(k)}$ from $DE_{\%}^{(k)}$.
   c. Compute $GE^{(k)}$, $EF_1^{(k)}$, $N_{\text{ex}}^{(k)}$, $VS^{(k)}$.
   d. Compute emission totals for all six pathways.
3. Sum across all sub-categories to obtain national pathway totals and total CO_2eq.
4. Record all outputs for iteration i.

After all iterations, compute summary statistics from the distribution of recorded outputs.

4.5 Recommended Iteration Count and Convergence

Monte Carlo estimates converge as $n_{\text{iter}} \to \infty$. For practical use:

- **1,000 iterations**: suitable only for a quick test run to verify the model runs without errors. Do not report results from 1,000 iterations.
- **10,000 iterations**: minimum recommended for any result intended for use. Convergence is

generally adequate for inventories with up to 5 sub-categories and 30 parameters.

- **25,000–30,000 iterations**: recommended for final reporting, especially when correlations are enabled or the inventory has many sub-categories or asymmetric distributions.
- **50,000 iterations**: high-precision run; use when the 95% CI bounds need to be stable to two significant figures.

Convergence check: Re-run with a different random seed (e.g. 123 and 456). If the 95% MoE changes by more than 1–2 percentage points, increase the number of iterations.

5 Uncertainty Metrics

The following metrics are reported for each emission pathway and for the total CO_2eq:

5.1 95% Confidence Interval

The 95% CI is defined as the 2.5th and 97.5th percentiles of the Monte Carlo output distribution:

$$\mathrm{CI}_{95} = [Q_{0.025},\ Q_{0.975}]$$

This is a simulation-based (quantile) interval: it does not assume a normal distribution and is therefore valid for asymmetric emission distributions.

5.2 Margin of Error (the IPCC reporting metric)

The margin of error (MoE%) is the half-width of the 95% CI expressed as a percentage of the mean:

$$\mathrm{MoE}\% = \frac{Q_{0.975} - Q_{0.025}}{2 \times \bar{x}} \times 100$$

MoE% is the metric IPCC Vol.1 Ch.3 Table 3.3 requires ("% uncertainty" in that table is defined as the half-width of the 95% CI divided by the mean). It is the headline number reported in the user-facing value boxes, in the IPCC Report tab, and in the Excel / Word downloads. Because MoE is derived from the empirical 2.5% / 97.5% quantiles of the Monte Carlo output, it correctly captures distribution skewness: for a perfectly normal distribution it reduces to $\approx 1.96 \times \mathrm{CV}\%$, but for skewed distributions (lognormal EFs, PERT-bounded fractions, manure-N_2O sums) MoE and $1.96 \cdot \mathrm{CV}$ differ.

5.3 Coefficient of Variation (kept for reference, not reported as headline)

The coefficient of variation (CV%) expresses uncertainty relative to the mean:

$$\mathrm{CV}\% = \frac{\sigma}{\bar{x}} \times 100$$

where σ is the standard deviation and $\bar{x}$ is the mean. CV% is **still computed internally** for every emission source (kept in the underlying `calc_all_uncertainty()` data frame for advanced reproducibility) but it is no longer surfaced in any user-facing display: IPCC Annex 7 expects MoE%, not CV%. It is undefined (reported as NA) when the mean emission is zero (e.g. pasture N_2O for fully housed systems).

5.4 IPCC Table 3.3 Format

IPCC (2006) Volume 1, Table 3.3 requires reporting uncertainty for each emission category as follows:

Source / Sink Category	Emission (t CH_4 or t N_2O)	AD Uncertainty (95% MoE %)	EF Uncertainty (95% MoE %)	Combined Uncertainty (95% MoE %)
Enteric fermentation CH_4	...	MoE% from AD-only run	MoE% from EF-only run	MoE% from combined run
...				

The tool generates this table automatically after each simulation run. The Excel download labels each uncertainty column explicitly as "(95% MoE %)" so there is no ambiguity with CV.

6 Uncertainty Decomposition

To separate the contributions of activity data (AD) uncertainty and emission factor (EF) uncertainty to the combined uncertainty, the tool performs three simulation runs:

1. **Combined run**: all parameters varied according to their distributions. This is the primary result.
2. **AD-only run**: emission factors fixed at their central values ($\sigma = 0$); only AD parameters vary. 95% MoE from this run = AD uncertainty contribution.
3. **EF-only run**: AD parameters fixed at their central values; only emission factors vary. 95% MoE from this run = EF uncertainty contribution.

This approach correctly handles multi-group inventories by running all three simulations through the full multi-sub-category equation chain and aggregating to national totals before computing metrics.

The decomposition reveals whether uncertainty reduction efforts should focus on improving animal population surveys (AD) or on refining emission factor estimates (EF).

Implementation note. The AD-only run pins ALL coefficient-side variance, not only the Parameters-sheet rows. The per-MMS sample matrices used for MCF, EF3, Frac_GasMS, Frac_LeachMS, and (when supplied) the MMS-allocation MS_s fractions are nulled out for the AD-only sub-run so iteration-to-iteration variation in those quantities cannot leak into the AD-only result. As a result, AD-only CV equals CV(N) for every emission source in a single-sub-category inventory and varies across sources only through the Herfindahl-like concentration of each source over multiple sub-categories.

7 Sensitivity Analysis

Sensitivity analysis identifies which input parameters have the greatest influence on the total emission estimate. The tool implements two complementary methods.

7.1 Standardised Regression Coefficients (SRC)

SRC is based on fitting a multiple linear regression of the standardised output on all standardised inputs:

$$Y^* = \beta_0 + \sum_j \beta_j X_j^* + \varepsilon$$

where $Y^* = (Y - \bar{Y})/\sigma_Y$ and $X_j^* = (X_j - \bar{X}_j)/\sigma_{X_j}$. The SRC for parameter j is the coefficient β_j from this regression.

Properties: - A positive SRC indicates that higher values of the parameter produce higher emissions. - The magnitude $|\beta_j|$ reflects the relative importance of parameter j. - SRC values range from approximately -1 to $+1$ when a single parameter dominates. - The R^2 of the regression indicates what fraction of the total variance in the output is explained by the linear model. High R^2 (> 0.9) indicates SRC is a reliable measure; low R^2 suggests non-linearity. - SRC requires only a single regression fit, making it computationally efficient for any number of parameters.

7.2 Tornado Chart Interpretation

Each parameter label on the tornado chart is formatted as `<parameter> (<sub_category>)`: for example `Ym (DINT_cow)`, `BW (DINT_heif)`, `MCF_solid_storage (DINT_cow)`. The sub-category in parentheses identifies which animal group each influential parameter belongs to so that reduction efforts can be targeted at the specific sub-category that drives uncertainty rather than at the parameter in aggregate. The same labels appear in the rank-correlation table and the downloadable Word / Excel sensitivity sheets.

The tornado chart displays the top parameters ranked by |SRC|:

- **Green bars** (positive coefficient): increasing this parameter increases total emissions; reduction in uncertainty would directly reduce total emission uncertainty.
- **Red bars** (negative coefficient): increasing this parameter decreases total emissions.
- **Dark (slate) bars**: IPCC-defined coefficients that cannot be improved by the national inventory compiler (e.g. GWP values, Cfi, Cp). Uncertainty in these parameters is irreducible without IPCC guidance changes.
- **Bright (green/red) bars**: user-reducible parameters where better national data could meaningfully reduce inventory uncertainty.

8 Correlation Structure

8.1 Activity Data Correlations

Correlations between activity data parameters (e.g. body weight and mature weight, milk yield and milk fat content) are entered in the Tab~4 *Activity-Data Correlations* card. Four modes are available; each is described in detail below.

- **No correlations** (independence): the default. All parameters are drawn independently. This is the conservative IPCC Approach 2 starting point and is appropriate when the inventory compiler has no defensible information about parameter co-movement.
- **From template (auto, time-series)**: the matrix is computed automatically from the `Parameter_TimeSeries` sheet of the upload using a **Spearman rank correlation** after detrending (first differences by default; linear and raw alternatives available). Spearman is preferred over Pearson because (i) it is robust to skewed marginals (Ym, Bo, milk yield), (ii) it matches the rank-correlation target the sampler reproduces (per IPCC Vol.1 Ch.3 §3.2.3.2), and (iii) it is the preferred option when the user wants the *rank* dependence preserved rather than a linear product-moment relationship that can be distorted by outliers. Requires ≥ 5 annual rows and ≥ 2 non-constant numeric columns to produce a usable matrix; otherwise the option is greyed out (see "UI gating" below).
- **Structural-defaults preset**: a small set of literature-based pairs drawn from the IPCC equation structure and the published livestock literature (see Section "Structural-Defaults Preset" below). Independent of the upload's time-series, so this is the option that works for single-year inventories without historical data.
- **Manual entry**: the user uploads a square symmetric CSV. The diagonal must be 1; entries are interpreted as Spearman rank correlations on $[-1, +1]$; the matrix is projected onto the nearest positive-definite correlation matrix if needed, with a warning if any entry shifts by more than 0.01.

UI gating. Modes whose prerequisites are missing are greyed out automatically and cannot be selected: `From template'' is disabled when the upload has no usable \texttt{Parameter\_TimeSeries} sheet,`Manual entry'' is disabled until the user uploads a CSV matrix, and `Structural defaults'' is disabled until a template (built-in example or custom upload) is loaded. The Run button additionally blocks the simulation with an explicit error if a no-op mode somehow slips through. These gates prevent the silent fall-back to`no correlations'' that would otherwise occur when a mode is selected without the data it needs.

Expected magnitude of effect on the headline. Picking the structural-defaults preset on a typical single-year inventory changes the 95% uncertainty interval by about one percentage point or less (built-in examples at 3,000 iterations: Country X 25.3% without correlations and 25.2% with the preset; Country Y 36.5% and 37.5%). An earlier version of the preset, with DE $\leftrightarrow$ Ym at -0.50,

widened the interval by 3–10%; the October 2026 revision lowered that pair to −0.20 (see the Structural-Defaults Preset section below). The central estimate is essentially unchanged because correlations preserve marginal means. The "From template (auto)'' mode produces effects in the same general range when the time-series is rich enough to deliver multiple correlated pairs; a sparse or near-static time-series will produce a much smaller effect (the matrix collapses to near-identity after detrending).

8.2 Emission Factor Correlations

Coefficients are sampled independently of one another. This is the standard IPCC Approach 2 assumption and, since October 2026, the tool's unconditional behaviour.

Earlier versions offered a block-structured mode in which coefficients were grouped into three blocks (energy-equation, manure-CH_4, manure-N), each carrying its own within-block ρ in $[0, 0.5]$, with cross-block correlation fixed at zero. The mode was withdrawn on two grounds. First, no inventory dataset supplies a within-literature correlation coefficient, so the control required an input the compiler could not evidence. Second, measurement showed most of it to be inert: on a 32-group inventory at 3,000 iterations, moving all three sliders from 0 to 0.5 moved the 95% margin of error from 9.09% to 11.00%, of which the energy block contributed −0.05 points and the manure-CH_4 block +0.05 points, both within sampling noise. The manure-N block accounted for the entire 1.9-point span, which follows from its parameter widths: EF3_PRP at ±221%, Frac_LEACH_PRP at ±150%, EF5 at ±89% and Frac_GASM_PRP at ±73%, against ≤ 30% for every coefficient in the other two blocks.

Correlation between coefficients remains expressible where it is evidenced. The unified correlation matrix spans both activity data and coefficients, so the structural-defaults preset (which includes the DE–Y_m pair) and a user-supplied manual matrix both act on coefficients. What was removed is the block abstraction and its sliders, not the capability.

8.3 Structural-Defaults Preset (Literature-Based)

The tool provides a **structural-defaults preset** correlation matrix as a middle ground between full independence and user-specified correlations. It assigns a non-zero rank correlation to seven pairs of parameters that are linked through the IPCC equations or through well-documented biology; all other pairs are zero. *The values are literature-based: the direction of each link comes from the IPCC equations and from published studies, and the size of each value was set by the project team from the evidence listed below. The IPCC publishes no numerical correlation coefficients for any livestock parameter pair.* Hover any non-zero cell in the heatmap for the per-pair source. The preset is appropriate when no empirical correlation data are available but the inventory compiler wants to avoid the unrealistic assumption of full independence.

The current preset contains seven pairs:

Pair	ρ	Biological reason	References and status
BW ↔ MW	+0.50	Live weight and mature weight enter the net energy for growth together (Eq 10.6, through $BW/(C \cdot MW)$), and animals of larger breeds are heavier at every age and have a larger mature weight. Set below the same-survey value (about 0.85) because MW is usually a breed-reference constant while BW comes from the census.	IPCC (2006) Eq 10.6; IPCC (2019) Table 10A.2 (r = 0.69 across 17 growing-cattle rows paired with the regional mature-cow weight). No published coefficient; project judgement.
BW ↔ WG	+0.40	Both enter Eq 10.6 ($NE_g = 22.02 \cdot (BW/(C \cdot MW))^{0.75} \cdot WG^{1.097}$); well-fed animals are both heavier and gain faster. The sign reverses for young stock approaching mature size.	Mehrban et al. (2021): phenotypic correlation of daily gain with body weight 0.49–0.63 in 15,279 Hanwoo cattle; IPCC (2019) Table 10A.2 growing categories 0.2–0.5. Literature-based.
Milk ↔ Fat	−0.30	Dilution effect: as milk yield rises, fat yield rises less than proportionally, so the fat percentage falls.	Boujenane (2002): −0.28 phenotypic in 20,140 Holstein-Friesian lactations; Nistor et al. (2011): −0.26 in 120 cows; IPCC (2019) Table 10A.1: −0.30 across nine regions. Literature-based.
Milk ↔ BW	+0.30	Larger dairy breeds and more intensive systems have both heavier cows and higher milk yields; within one breed the link is weak.	IPCC (2019) Table 10A.1: 0.84 across regions; Berry and Evans (2022): the heaviest quarter of 16,980 Holstein-Friesian cows yields only 3–4% more; Berry et al. (2003): genetic correlation close to zero. Project judgement between the across-breed and within-breed cases; a single-breed herd may use 0.10–0.20.
Milk ↔ DE	+0.30	More digestible feed gives more net energy per kg of feed and supports higher milk yield.	NRC (2001) Ch. 2; IPCC (2019) Table 10A.1: 0.65 across regions; no within-system coefficient is published. Project judgement, conservative.

Pair	ρ	Biological reason	References and status
DE ↔ CP	+0.20	Within one forage, digestibility and crude protein fall together as the plant matures. Across different feeds the link is weak, because protein and digestibility depend on different things (plant family, plant part, processing).	INRAE-CIRAD-AFZ (2018): $r = -0.07$ over 193 plant feed materials, and energy digestibility 57–64% across the dehydrated-alfalfa protein grades; Feedipedia: 0.1–0.2 over 82 forages with measured digestibility. Project judgement near the measured across-feed value; use 0.3–0.5 when the uncertainty is mainly about forage maturity.
DE ↔ Ym	−0.20	IPCC Table 10.12 assigns a lower Ym to more digestible diets, mainly because high digestibility goes with high feeding level and grain-rich rations, which shift rumen fermentation and speed up passage; within forage-based diets, methane per unit of energy changes little with digestibility or rises slightly.	IPCC (2019) Table 10.12 (implies $r \approx 0$ under the Country X ranges, −0.32 read as a straight line); Ramin and Huhtanen (2013): positive link on forage diets (298 diets); Johnson and Johnson (1995). System-dependent: −0.20 for forage-based systems; −0.3 to −0.5 for high-grain dairy or feedlot systems via the manual matrix.

Revision history. May 2026 audit: a previously included N ↔ BW = 0.30 pair was removed (N is sampled as a scalar broadcast, so the pair was inert), and Cfi ↔ Ca was lowered from 0.60 to 0.30. June 2026 livestock-science review: BW ↔ MW lowered from 0.85 to 0.50 (mixed-source provenance); Milk ↔ BW and Milk ↔ DE added as cross-group biological linkages; Milk ↔ pct_pregnant dropped (the sign is ambiguous between per-cow genetic and per-herd management interpretations) and Cfi ↔ Ca dropped (independent inputs in Eq 10.3 / 10.4, not jointly estimated). October 2026 evidence run: the label changed from "expert-elicited" to "literature-based"; DE ↔ CP lowered from +0.50 to +0.20, because the previously cited "cross-feedstuff correlations of 0.4–0.7" appear in none of the named sources and measured feed tables show no across-feed link; DE ↔ Ym lowered from −0.50 to −0.20, because the previously cited Eq 10.21 is the emission-factor equation rather than a Ym–DE relation, Table 10.12 implies a near-zero correlation under typical input ranges, Niu et al. (2018) does not use digestibility as a predictor, and measured forage-diet data give a positive link; citations that could not be verified (Brody 1945; Wilmink 1987; VandeHaar 1998; Stallings and Knowlton 2013; the "NRC 2001 joint regression") were removed from all seven sources. The resulting matrix is positive-definite without nearest-PD repair (smallest eigenvalue 0.26).

8.4 Cross-Block Correlations

The preset and the manual / time-series matrices support cross-block correlations (i.e. correlations between an activity-data parameter and a coefficient: for example BW ↔ Cfi). This is implemented via a single unified correlation matrix that spans the union of all parameter names; the sampler draws

from one multivariate distribution covering both blocks rather than two independent block-level distributions. Independence within and across blocks is the default.

8.5 Time-Series Correlation Matrix

When the user selects "From template'' with a `Parameter_TimeSeries` sheet loaded, the tool starts from the seven structural pairs of the previous section and lets the country's own series set the size of a pair, under three conditions: the series has at least ten years; the Spearman rank correlation of the year-to-year changes of the two parameters (first differences, which remove the shared long-run growth that is not the joint uncertainty Approach 2 is meant to capture) is different from zero at $p < 0.05$; and its sign agrees with the literature. A pair that fails a condition keeps its literature value. No other pair is created. The matrix is then projected onto the nearest positive-definite correlation matrix.

The restriction (30 October 2026) replaces an earlier behaviour that applied a correlation between every two columns of the sheet. Two reasons. With n years there are $n - 1$ changes per parameter, and with five years any pair can show a rank correlation near ± 0.9 by chance. And the co-movement of national annual values measures shared drivers (weather, policy, data revisions), which is not the correlation between estimation errors that IPCC Approach 2 asks for; the literature pairs describe the mechanism, which is closer to it. The Correlations tab shows the check pair by pair: literature value, value in the series, number of years, p-value, and which one the run uses. When no pair passes, the run uses the structural defaults and the tab says so.

8.6 Across-Group Coefficient Correlation

An inventory is normally disaggregated into groups (cattle type × production system × animal sub-category). A second decision then arises, exactly parallel to the year-to-year decision described in the next section: within a single Monte Carlo iteration, when two groups use the *same* value for a coefficient, should they receive the same draw or separate draws?

The question is not cosmetic. Uncertainty in a shared estimate is **systematic**. If the true Y_m is 7.0 rather than the 6.5 applied to every group, every group is wrong in the same direction, by the same relative amount, simultaneously; nothing cancels in the sum. Drawing it independently per group asserts instead that the published figure is re-drawn afresh for each sub-category, and the resulting errors partially cancel. Because the cancellation goes as $1/\sqrt{N}$, the reported uncertainty then falls as the inventory is split more finely.

The magnitude is large. Holding the herd, the parameters and the uncertainty inputs fixed and varying only the number of groups:

Groups	Mean (t CO_2eq)	95% MoE
1	1,188,109	25.1%
4	1,189,861	12.5%
16	1,188,003	6.3%
64	1,188,164	3.1%
251	1,188,033	1.5%

The mean is stable throughout; only the uncertainty moves, and it moves purely with a reporting choice.

IPCC 2019 Refinement Vol.1 Ch.3 addresses this directly. It warns that subcategories `can be highly correlated, because either the AD are derived from the same source or the EFs have parameters in common''; it states that`in the case of full correlation among categories, aggregation of these categories is the recommended procedure'‘; and it recommends Approach 2 `where partial correlations are known to exist and are relevant''. It also gives the general rule, in the context of the time dimension:`emission factors (and other estimation parameters) uncertainties will tend to be correlated between years while activity data will tend to be uncorrelated between years.''

Status in the app (30 October 2026). Every run uses the shared draws described below, in single-year and in trend mode; there is no setting. The rule is stated on the Correlations tab and in the user guide. (Between 28 and 30 October the app had sampled every group independently.)

The rule the tool applies when shared draws are used. Groups carrying an identical estimate for a coefficient share a single draw; activity data stays independent. Formally, for each coefficient the groups are partitioned by their (mean, lower bound, upper bound, distribution) tuple; every group within a partition receives the same per-iteration draw, and separate partitions are drawn independently. Population N is never shared.

Three properties follow:

- **Shared IPCC defaults are handled automatically.** Every group that fell back to the catalogue carries the identical value, forms one partition, and is therefore fully correlated, which is the only defensible treatment of a single published number.
- **Block structures are handled by the same mechanism.** An inventory with one value per agro-region forms one partition per region: correlated within a region, independent between regions, with no configuration.
- **It is self-correcting.** A compiler who genuinely estimated a coefficient separately per group has entered different numbers, so those groups land in different partitions and remain independent. An identical value is the evidence of a shared source.

Partitioning on the full tuple also guarantees that the marginal distributions within a partition are

identical, which is the precondition for sharing sampled values directly rather than sharing ranks.

Shared draws and the correlation matrix (30 October 2026). A shared draw is one number given to every group in its partition, so it cannot follow a different partner draw in each of those groups: one Y_m handed to twelve groups cannot be low where a group's own DE is high and high where it is low. The active correlation matrix is therefore applied between shared estimates only where they cover exactly the same set of groups (one DE and one Y_m for the same twelve groups move together as the matrix says, with the same restricted-pairing procedure and unchanged marginals), and between a group's own values among themselves, as before. A pair made of a shared estimate and a group-specific value is left uncorrelated. In trend mode the same rule is applied to the coefficient blocks drawn for the whole series (`fully correlated'') and to the normal innovations of the AR(1) chain` (partial'), so the matrix reaches the series as well as the single-year run.

Verification. Because IPCC states that full correlation should be equivalent to aggregation, the implementation has a hard acceptance property: splitting an inventory into more groups must not change the reported uncertainty. With sharing enabled the table above becomes approximately flat (25.1% on one group, 22.7% on 251). The residual decline is the genuinely independent activity-data component averaging out, and is quantitatively consistent with the 10% uncertainty placed on N in that test: $\sqrt{25.1^2 - 22.7^2} \approx 10.7\%$.

A real inventory. The same comparison on the Zambia Tier 2 inventory (32 groups, 800 parameter rows, 4,000 draws, no activity-data correlations). The mean is unchanged (9.25 Mt CO_2eq); the activity-data-only margin is identical (12.9%) because population is never shared; the whole difference sits in the emission-factor side, where Ym (one value in 12 groups), Cfi (one value in 22) and the pasture N_2O factors (one value in all 32) are drawn once instead of 32 times. Cutting every group into two and four identical parts, same animals and N divided, reproduces the Country X pattern.

	Separate draws	**Shared draws**
Total CO_2eq, 95% MoE (32 groups)	23.7%	32.5%
Enteric CH_4	24.3%	33.4%
Pasture N_2O, direct / indirect	39.7% / 58.2%	103% / 148%
AD-only / EF-only	12.9% / 19.2%	12.9% / 29.2%
Same herd in 64 groups	16.8%	30.9%
Same herd in 128 groups	11.5%	30.3%

In the sensitivity ranking the shared estimates appear once each, labelled with the value and the groups they cover ("DE = 52.8 (extensive beef, 6 groups)''); with separate draws the same uncertainty is spread over one bar per group and the ranking is dominated by the largest groups.

Scope. Only full sharing is implemented. Partial sharing ($0 < \rho < 1$), appropriate where a shared systematic component coexists with independent per-group sampling error, is not offered: it requires

an $n_{\text{iter}} \times n_{\text{groups}}$ matrix per parameter held simultaneously, which is not feasible within the hosting memory budget at the group counts where the issue matters most. IPCC positions partial correlation as a refinement to apply "where known to exist and relevant''.

8.7 Animal Numbers Between Groups and Between Years

Population N is never shared between groups by the rule of the previous section (each group's own count is its own estimate), yet in most inventories the groups' numbers are shares of one national count, and in a time series the same count, interpolated, serves many years. Drawing each group's N on its own, afresh every year, lets a census error cancel between groups and between years. IPCC 2019 Refinement Vol.1 Ch.3 names activity data "derived from the same source'' as a case of correlated sub-categories, and the 2006 Guidelines present the independence of activity data between years (Approach 1, Eq. 3.2) as a simplifying assumption that Approach 2 can relax where the same source serves both years.

Since 30 October 2026 the Correlations tab asks two questions about N and the sampler honours them. Normal scores z (one column per group) carry the structure: between groups, $z_g = \sqrt{\rho}\, c + \sqrt{1-\rho}\, \varepsilon_g$ with a common c and $\rho = 1$ (one count split into groups, the preselected answer), 0.5 (partly) or 0 (separate counts); between years, in trend mode, the scores are kept for every year (one census, interpolated), follow an AR(1) with $\rho = 0.7$ (a census every few years) or are renewed each year (annual survey, the preselected answer). Each group's own marginal distribution is then placed on the ranks of its column, so means and ranges are unchanged; only the way the draws are tied together changes. The EF-only run of the decomposition drops these draws, as it holds N at its mean. In the sensitivity ranking a count split into groups appears as one bar ("N (one count, 32 groups)''), the sum of the groups' draws, because their ranks are identical and a regression cannot separate them.

Zambia (32 groups, no activity-data matrix, shared coefficient draws). Single year, 4,000 draws: total 95% MoE 32.5% with separate counts, 40.1% partly, 47.1% with one count split; AD-only 12.9%, 27.1%, 36.5%; EF-only 29.2% throughout; the mean unchanged. Trend 1990 to 2024, 2,000 draws per year, coefficients fully correlated between years:

Between groups	Between years	1990 MoE	2024 MoE	95% range of the trend (pp)
separate counts	annual survey	34.8%	35.7%	−61 to +78
separate counts	census every few years	35.2%	35.7%	−62 to +82
separate counts	one census, interpolated	35.2%	35.4%	−3 to +2
partly	annual survey	48.1%	48.8%	−110 to +201
partly	one census, interpolated	48.1%	47.7%	−3 to +4
one count split	annual survey	58.3%	58.6%	−140 to +315
one count split	census every few years	58.3%	59.5%	−138 to +314
one count split	one census, interpolated	58.3%	57.9%	−3 to +4

The between-groups choice moves each year's margin; the between-years choice moves the trend's: a census error kept for every year cancels in the comparison of two years, a renewed one does not.

8.8 Year-to-Year Coefficient Correlation (Trend Mode)

When the tool is run in **Trend mode** (multi-year inventory), an additional decision arises: within a single Monte Carlo iteration, how should the IPCC coefficients (Cfi, Ca, Ym, Bo, MCF, EF3, EF4, EF5, ...) move from one year to the next? IPCC Volume 1 Chapter 3 Section 3.2.3.2 (Trend Analysis) and 2019 Refinement §3.2.2.4 discuss this explicitly: emission factor uncertainties are typically estimated once for an inventory series and reused across years (correlated), while activity data is normally re-estimated each year and treated as independent across years.

The tool exposes three options on the Simulate tab when Trend mode is selected. Activity data (N, BW, WG, Milk, ...) is re-sampled fresh each year regardless of this setting:

Option	What happens inside one Monte Carlo iteration
Fully correlated (default)	The coefficient draw for year t is the *same* as for year 0. If Ym is drawn at 6.2% for year 0, it is 6.2% for every other year in that iteration. The trend then reflects activity-data changes only: coefficient uncertainty cancels when subtracting two years. Matches IPCC 2019 §3.2.2.4 default behaviour and the Ogle et al. (2003) "identical random seed" construction cited in V1 Ch3 Box 3.2.
Partial (AR(1), $\rho = 0.7$)	Coefficients drift slowly between years with a rank correlation of $\rho^{\lvert i-j\rvert}$ between year i and year j ($\rho = 0.7$). Independent samples are drawn for each year from the coefficient's marginal and then reordered across years so the realised Spearman rank correlation matches the AR(1) target: the same restricted-pairing procedure used within a year (per IPCC Vol.1 Ch.3 §3.2.3.2), applied across years. Neighbouring years share most of the same observational basis but the coefficient can move over a decade. Appropriate when the EF is re-estimated periodically (e.g. every 3–5 years).
Independent	Each year gets a fresh, independent coefficient draw. Maximises the EF contribution to trend uncertainty. Realistic only when every emission factor is genuinely re-measured with new field data every year, which is uncommon for national inventories.

The choice is implemented in `R/trend_tab.R`. The `full'' branch pre-samples a single $n_{\text{iter}} \times n_{\text{coef}}$ matrix and reuses it for every year. The`partial'' branch builds an $n_{\text{iter}} \times n_{\text{years}}$ rank-correlated draw for each coefficient against an AR(1) target rank correlation matrix, then runs each year's MC with the corresponding column slice. The "independent'' branch re-samples coefficients inside every per-year call.

Draws per year, and the two reported years. The series runs every year at the trend iteration count (2,000 by default), which is enough for the chart and the slope: on a national inventory of 32 sub-categories the 95% margin of error of a year is known to about ±1 percentage point at 2,000 draws against ±0.6 at 10,000, and the standard error of the mean is 0.3% against 0.13%. The headline figures and IPCC Table 3.3 depend on the base year and year t only, so after the series those two years are simulated again, as a pair, at the reporting draw count (10,000 by default, the single-year setting): the same coefficient draws in both years under `fully correlated'', the lag-$k$ AR(1) link under`partial'', independent under "independent'', each year with the

seed the series gives it. The activity-data and emission-factor columns of Table 3.3 (two further simulations for year t with the other side held at its mean) are run at the same count on the same systems. Run at the series' own count, the paired run reproduces the series' draws for the two years exactly, so it is the same computation at higher precision. Choosing two other years on the report tab runs them the same way.

Direction of effect. For an emission factor whose uncertainty is largely systematic, full correlation produces the *smallest* trend uncertainty (because the EF cancels) and independent sampling the *largest*. IPCC V1 Ch3 p. 25 explicitly notes that `positive correlations in uncertainties when comparing two years as part of trend analysis will decrease uncertainty in the trend''. Choosing`independent'' when the EFs are in fact reused will over-state trend uncertainty; choosing `full'' when the EFs are in fact re-measured each year will under-state it. The IPCC default` (full'') is the conservative choice for the typical case where the inventory compiler is using IPCC defaults or a single national estimation programme.

9 Validation and Quality Assurance

The tool includes a QA/QC module (`R/utils_qaqc.R`) that runs a series of checks against the loaded parameter data before simulation. Each check returns one of five statuses:

Status	Trigger	What the user sees
Pass	All structural checks pass. For `BW`, the mean is also within ±50% of the continental IPCC default (Annex Tables 10A.1 / 10A.2 / 10A.3).	Green tick; no action needed.
Warn	For `BW`: value deviates 50–200% from the continental IPCC default. For any parameter: implausible bounds ($BW > 1500$ kg, $Y_m > 12\%$); a fractional parameter is using an unbounded distribution; or another non-fatal structural condition fired.	Amber row; the message either suggests a fix or asks the user to document the country-specific deviation.
Fail	For `BW`: value deviates more than 200% from the continental IPCC default. For any parameter: $DE_{\%} \leq 0$ or ≥ 100; or $lower > upper$; or an MMS fraction-sum is more than 1% off 100%; or a Beta is configured with mean outside $(0, 1)$.	Red row; simulation will still run but the result will be flagged in the report.
Missing	A core parameter was not supplied in the upload; auto-filled with the IPCC default. Applies to every parameter, not just `BW`.	Grey row; the message names the IPCC table or equation used and, for context-dependent parameters (EF3_PRP, EF4, Frac_LeachPRP, Bo, MCF, Ym, Cfi, Ca), appends an explicit hint of which alternative IPCC table row should be reviewed for the inventory's climate / production system / animal class.
Info	Reserved for future advisory messages.	Blue row.

The Pass / Warn / Fail thresholds for benchmark deviation apply **only to body weight (`BW`)**, the single parameter for which the tool holds a defensible continental IPCC default lookup: Vol.4 Ch.10 Annex Table 10A.1 (dairy cows), Annex Table 10A.2 (other cattle), or Annex Table 10A.3 (buffalo). Earlier versions of the tool also flagged deviations on Milk, MW, DE, Ym, and Bo against heuristic mid-points derived from the same Annex 10A illustrative tables; these were removed (Andreas Wilkes review, May 2026) because those mid-points were not directly citable to a published IPCC default and a reviewer auditing the QA tab could not locate them in the guidelines. The `ipcc_default` column of `PARAM_CATALOGUE` is still used to auto-fill missing values and is sourced from Vol.4 Ch.10 (Tables 10.4, 10.5, 10.7, 10.11, 10.12, 10.16(a), 10.17, 10.21, 10.22, 10.23) and Vol.4 Ch.11 (Tables 11.1, 11.3), it is just no longer used as a deviation benchmark for parameters other than BW. Reinstating per-parameter benchmarks properly would require a multi-dimensional table indexed by continent, IPCC version (2006 vs 2019R), production system, and animal sub-category; that is deferred to a future release. Auto-filled parameters are listed in a banner on Tab~1 immediately after data upload; they should be replaced with country-specific values wherever possible, because both the central estimate and the propagated uncertainty depend on them.

10 Verification of Calculation Accuracy

The previous section describes the checks the tool applies to the user's *inputs*. This section describes how the tool's *calculation engine*: the implementation of the IPCC equation chain and the Monte Carlo sampler, has itself been verified to produce correct numbers. Input validation and calculation verification are complementary: the former guards against bad data going in, the latter guarantees the equations are implemented correctly.

10.1 Equation-chain verification against manual calculation

The core of the verification is a fixed, fully worked test case. For a single representative animal with a complete set of fixed parameters, every intermediate quantity in the IPCC Tier 2 chain was computed independently by hand and in a spreadsheet, directly from the published equations: the net-energy terms (NE_m, NE_a, NE_g, NE_l, NE_w, NE_p), the energy ratios (REM, REG), gross energy intake (GE), the enteric emission factor (EF_1), volatile-solid excretion (VS), nitrogen excretion (N_{ex}), and all four manure- and pasture-N_2O pathways, through to the inventory totals (total CH_4, total N_2O, and total CO_2eq under each GWP basis).

The tool's computed value for each of these quantities must match the independently calculated reference to a relative tolerance of 1×10^{-4}. These comparisons are encoded as automated assertions that run on every build, so any future change to the calculation code that altered a result: even slightly, would be caught immediately. This is the primary guarantee that the equations in Section "IPCC Tier 2 Emission Equations" are implemented exactly as written.

10.2 Cross-comparison with an independent Monte Carlo package

Beyond the single worked case, the tool has been exercised on real Tier 2 national cattle inventories and its results compared against two independent references: (i) manual re-calculation of the per-head emission factors and inventory totals, and (ii) an independent commercial Monte Carlo package configured with the same parameters, distributions and iteration count.

On these real inventories the tool reproduced the deterministic emission totals from the manual calculation, and produced a 95% uncertainty interval consistent with the independent Monte Carlo package to within the sampling variability expected between two stochastic runs. %% OPTIONAL: if you wish to quote a specific figure, insert it here, e.g. %% "(agreement within X% on the 95% margin of error for the manure-N2O pathway)". A regression guard retained in the automated test suite keeps the manure-N_2O pathway within the range established by this cross-comparison, so a future code change cannot silently introduce a large (e.g. order-of-magnitude) error in that pathway.

10.3 Automated regression test suite

The two verification activities above are part of a broader automated test suite (91 checks at the time of writing) that is run before each release and deployment. The checks are organised into seven groups:

- **A: IPCC equation chain:** every intermediate and total against the hand-calculated reference case (Section "Equation-chain verification" above).
- **B: Monte Carlo sampler:** empirical means of the sampled marginals match their analytical means; realised Spearman rank correlations match the requested targets; the AR(1) year-to-year correlation structure behaves as specified.
- **C: Application routes:** emission-source filtering (all sources, enteric only, manure CH_4 only, N_2O only), the three GWP bases (AR4 / AR5 / AR6), the correlation modes, and the AD / EF / combined uncertainty decomposition.
- **D: Trend (multi-year) pipeline:** the full / partial / independent year-to-year coefficient-correlation modes produce valid multi-year results.
- **E: Multi-sub-category aggregation:** per-sub-category emissions sum correctly to national totals and scale as expected with population.
- **F: Edge cases and validators:** missing-value handling, manure-management fraction bounds, lower/upper-bound consistency, zero-population handling, and the multi-system N_2O calculations.
- **G: Outputs:** the Excel and Word/CSV downloads and the correlation-matrix round-trip produce well-formed, internally consistent files.

The full multi-sub-category inventory chain must pass all checks before the tool is published.

10.4 Verification of IPCC default values and uncertainty ranges

Every IPCC default value and uncertainty range pre-loaded in the tool: the energy-equation coefficients, the methane conversion factor Y_m, the maximum methane producing capacity B_o, the manure-system MCF values, and the N_2O emission factors and volatilisation / leaching fractions (EF_3, EF_4, EF_5, FracGASM, FracLEACH), has been checked line-by-line against the IPCC 2019 Refinement source text (Volume 4, Chapter 10 and Chapter 11). This review confirmed the central values and corrected two pre-loaded uncertainty ranges (B_o and the pasture direct-N_2O factor $EF_{3,\mathrm{PRP}}$) to match the climate-specific ranges in Tables 10.16a and 11.1.

10.5 Default uncertainty ranges

The tool applies a default uncertainty range only where the literature gives one that applies to any country. These are the ranges IPCC publishes with its default values, and the expert-judgement ranges of Monni et al. (2007) for four IPCC equation coefficients that no country can measure; the UK agriculture inventory uses the same four ranges (Milne et al., 2014). Table 5 lists them.

Every other input describes the country's own data: animal numbers, weights, weight gain, milk yield and fat, pregnancy, feed digestibility and crude protein, MCF, and the remaining manure inputs. Its range reflects how those data were collected, so it must come from the country. IPCC 2019, Vol. 1, Table 3.6 gives ranges for some of these inputs, but for one national example (Italy);

Monni et al. (2007) give them for Finland's animal register. Neither applies to another country's data. When the country gives no range for such an input, the tool uses a placeholder so that the simulation can run: the values the tool used before October 2026 for the parameters, and for the manure inputs MCF ±25%, EF_3 a factor of 2, and the volatilisation and leaching fractions ±50%. A placeholder has no published source. It is reported as a warning on the QA/QC tab, listed in the Word report, and marked in the Excel and CSV exports, so that it is never mistaken for a default or for the country's own range.

The asymmetric IPCC ranges belong to the IPCC default value. They are applied only when the input carries that value; a country that uses its own value supplies its own range. The percentage ranges are applied around the value given, and the QA/QC tab notes each one so that a country with its own value can replace it.

Table 5: Inputs with a default uncertainty range.

Input	Default range	Source
Y_m	±20%	IPCC 2019, Vol. 4, Table 10.12, footnote 3
B_o	±15%	IPCC 2019, Vol. 4, Table 10.16A
$EF_{3,\mathrm{PRP}}$	0.0005 to 0.027	IPCC 2019, Vol. 4, Table 11.1 (lower bound 0 in IPCC; a small positive floor is used)
EF_4	0.011 to 0.017	IPCC 2019, Vol. 4, Table 11.3
EF_5	0.0005 to 0.020	IPCC 2019, Vol. 4, Table 11.3
$\mathrm{Frac}_{\mathrm{GASM}}$ (pasture)	0.005 to 0.31	IPCC 2019, Vol. 4, Table 11.3
$\mathrm{Frac}_{\mathrm{LEACH}}$ (pasture)	0.01 to 0.73	IPCC 2019, Vol. 4, Table 11.3
EF_3 by manure system	a factor of 2	IPCC 2019, Vol. 4, Table 10.21, footnote 24
$\mathrm{Frac}_{\mathrm{GasMS}}$, $\mathrm{Frac}_{\mathrm{LeachMS}}$ by system	the Table 10.22 range, where IPCC gives one	IPCC 2019, Vol. 4, Table 10.22
C_{fi}, C_a, C	±30%	Monni et al. (2007), Table I
C_p	±10%	Monni et al. (2007), Table I

10.6 Reproducibility

Each simulation is driven by an explicit random seed, so any run can be reproduced exactly by re-supplying the same inputs and seed. The convergence guidance in Section "Recommended Iteration Count and Convergence": re-running with a second seed and checking that the 95% margin of error is stable, lets a user confirm that a reported result is converged rather than an artefact of a particular random draw.

11 Reporting

The results calculated by the tool are designed to align with IPCC uncertainty reporting guidelines (IPCC 2006 Vol 1 Ch 3). The outputs reported by the tool can be used in a hybrid Approach 1 / Approach 2 uncertainty analysis for the whole inventory (IPCC 2006 §3.2.3.3) by entering into Table 3.3 the results reported by the tool for cattle emission sources alongside Approach 1 results for other emission categories. The outputs produced by the tool can also be used as inputs to a full Approach 2 analysis of uncertainty for the whole inventory.

11.1 IPCC Table 3.3 Inputs

In the case of a hybrid Approach 1 / Approach 2, IPCC 2006 Table 3.3 (see also the IPCC 2019 MS Excel tool for Approach 1, and the IPCC 2019 Vol 1 Ch 3 addendum) requires the following inputs to characterise uncertainty of each emission source, each expressed as $\pm x\%$ of the inventory emission estimate:

- Activity-data (population) uncertainty (Column E);
- Emission-factor uncertainty (Column F);
- Combined uncertainty (Column G).

The tool reports all three figures for each cattle emission source via the IPCC Report tab and the Excel / Word downloads. The activity-data vs emission-factor split follows the convention adopted in this tool: AD = animal population (N) only; coefficient (EF) = the IPCC equation parameters that combine into the per-head emission factor (Cfi, Ca, Ym, Bo, MCF, EF3, EF4, EF5, Frac_GasMS, Frac_LeachMS, Frac_PRP, EF3_PRP, Frac_GasPRP, Frac_LeachPRP).

11.2 UNFCCC Common Reporting Tables Category Map

The tool can calculate AD, EF and combined uncertainties for one or more cattle categories. While country-specific classifications are allowed for, the UNFCCC Common Reporting Tables (CRTs) indicate reporting of emissions from cattle at the following levels of disaggregation:

- **3.A Enteric fermentation (CH_4)**
  - 3.A.1.a Dairy cattle
  - 3.A.1.b Non-dairy cattle
- **3.B Manure management** (CH_4; direct and indirect N_2O combined)
  - 3.B.1 Cattle (3.B.1.a Dairy, 3.B.1.b Non-dairy)
- **3.D Agricultural soils** (direct and indirect N_2O)
  - 3.D.1.c Urine and dung deposited by grazing animals (direct N_2O)
  - 3.D.2.a Atmospheric deposition (multiple sources, including dung and urine deposited on pasture)
  - 3.D.2.b Nitrogen leaching and run-off (multiple sources, including dung and urine deposited on pasture)

Country-specific options in the CRTs may include reporting of emissions at the level of production systems, regions or cattle sub-categories.

11.3 Disaggregation Levels

To facilitate uncertainty analysis at these different levels of disaggregation, the default outputs in the tool are produced at three disaggregation levels concurrently:

- **Level 1**: disaggregated by cattle type (dairy, non-dairy), emission source (3.A, 3.B, 3.D) and gas (CH_4, N_2O). Available in the IPCC Report tab and the headline summary of the Word / Excel run-summary.
- **Level 2**: Level 1 plus a user-defined second level (production system, region) using the template's `cattle_type` and `region` fields. Available in the per-cattle-type results section of the Word run-summary.
- **Level 3**: Level 2 plus cattle sub-categories (dairy_cow, heifer, bull, etc.). Available in the per-(cattle type × emission source) breakdown of the Word run-summary and the per-sub-category CSV download from the IPCC Report tab.

A user-selectable toggle to choose the reporting disaggregation level for the whole tool is not implemented in this release; the three levels are produced concurrently in their respective output channels. A single-toggle option may be added in a future version after feedback from UNFCCC partners on the right default level.

12 References

Berry, D.P., and Evans, R.D. (2022). The response to genetic merit for milk production in dairy cows differs by cow body weight. *JDS Communications*, 3(1), 32–37. https://doi.org/10.3168/jdsc.2021-0115

Berry, D.P., Buckley, F., Dillon, P., Evans, R.D., Rath, M., and Veerkamp, R.F. (2003). Genetic relationships among body condition score, body weight, milk yield, and fertility in dairy cows. *Journal of Dairy Science*, 86(6), 2193–2204.

Boujenane, I. (2002). Estimation des param‘etres g'en'etiques et ph'enotypiques de la production laiti‘ere des vaches de race Frisonnes Holstein au Maroc. *Revue d''elevage et de m'edecine v'et'erinaire des pays tropicaux*, 55(1), 63–67. https://doi.org/10.19182/remvt.9848

Feedipedia (INRAE, CIRAD, AFZ and FAO). Animal feed resources information system: nutritional tables of forage products. https://www.feedipedia.org (forage datasheets consulted October 2026).

Frey, H.C., and Rhodes, D.S. (1998). Characterizing, Simulating, and Analyzing Variability and Uncertainty: An Illustration of Methods Using an Air Toxics Emissions Example. *Human and Ecological Risk Assessment*, 4(4), 762–797.

IPCC (2006). 2006 IPCC Guidelines for National Greenhouse Gas Inventories. Volume 1: General Guidance and Reporting, Chapter 3: Uncertainties. Eggleston, H.S., Buendia, L., Miwa, K., Ngara, T., and Tanabe, K. (Eds). IGES, Japan.

IPCC (2006). 2006 IPCC Guidelines for National Greenhouse Gas Inventories. Volume 4: Agriculture, Forestry and Other Land Use, Chapter 10: Emissions from Livestock and Manure Management. IGES, Japan.

IPCC (2019). 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories. Volume 4: Agriculture, Forestry and Other Land Use, Chapter 10 (Emissions from Livestock and Manure Management) and Chapter 11 (N_2O Emissions from Managed Soils, and CO_2 Emissions from Lime and Urea Application). Calvo Buendia, E., Tanabe, K., Kranjc, A., et al. (Eds). IPCC, Switzerland.

IPCC (2019). 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories. Volume 1: General Guidance and Reporting, Chapter 3: Uncertainties. IPCC, Switzerland.

Iman, R.L., and Conover, W.J. (1982). A distribution-free approach to inducing rank correlation among input variables. *Communications in Statistics: Simulation and Computation*, 11(3), 311–334. (The restricted-pairing algorithm explicitly cited in IPCC 2006 Vol.1 Ch.3 §3.2.3.2 and used by this tool whenever a Spearman correlation matrix is estimated from a time series.)

INRAE, CIRAD, and AFZ (2018). Tables of composition and nutritional values of feed materials (feedtables.com): energy digestibility in ruminants and crude protein. https://www.feedtables.com (consulted October 2026).

Johnson, K.A., and Johnson, D.E. (1995). Methane emissions from cattle. *Journal of Animal Science*, 73(8), 2483–2492.

Jones, D.B. (1931). Factors for Converting Percentages of Nitrogen in Foods and Feeds into Percentages of Proteins. *USDA Circular* No. 183. United States Department of Agriculture, Washington, D.C.

Karimi-Zindashty, Y., MacDonald, J.D., Nitschelm, J., Del Grosso, S.J., and Desjardins, R.L. (2012). Sources of uncertainty in the Canadian national livestock greenhouse gas inventory. *Journal of Agricultural Science*, 150(6), 677–690.

Mehrban, H., Naserkheil, M., Lee, D., and Ib'a~nez-Escriche, N. (2021). Genetic parameters and correlations of related feed efficiency, growth, and carcass traits in Hanwoo beef cattle. *Animal Bioscience*, 34(5), 824–832. https://doi.org/10.5713/ajas.20.0135

Milne, A.E., Glendining, M.J., Bellamy, P., Misselbrook, T., Gilhespy, S., Rivas Casado, M., Hulin, A., Van Oijen, M., and Whitmore, A.P. (2014). Analysis of uncertainties in the estimates of nitrous oxide and methane emissions in the UK's greenhouse gas inventory for agriculture. *Atmospheric Environment*, 82, 94–105.

Monni, S., Per"al"a, P., and Regina, K. (2007). Uncertainty in agricultural CH_4 and N_2O emissions from Finland: possibilities to increase accuracy in emission estimates. *Mitigation and Adaptation Strategies for Global Change*, 12, 545–571.

National Research Council (2001). *Nutrient Requirements of Dairy Cattle*, 7th revised edition. National Academies Press, Washington, D.C. (Chapter 2, Energy.)

Nistor, E., Bampidis, V., Matiuti, M., Pentea, M., Pet, L., and Ciolac, V. (2011). Estimates of phenotypic parameters for milk production traits in Romanian Black Spotted breed dairy cows. *Scientific Papers Animal Science and Biotechnologies*, 44(1), 316.

Penman, J., Kruger, D., Galbally, I., Hiraishi, T., Nyenzi, B., Emmanuel, S., Buendia, L., Hoppaus, R., Martinsen, T., Meijer, J., Miwa, K., and Tanabe, K. (2000). *Good Practice Guidance and Uncertainty Management in National Greenhouse Gas Inventories*. IPCC/OECD/IEA/IGES, Hayama, Japan.

Ramin, M., and Huhtanen, P. (2013). Development of equations for predicting methane emissions from ruminants. *Journal of Dairy Science*, 96(4), 2476–2493. https://doi.org/10.3168/jds.2012-6095

Saltelli, A., Ratto, M., Andres, T., Campolongo, F., Cariboni, J., Gatelli, D., Saisana, M., and

Tarantola, S. (2008). *Global Sensitivity Analysis: The Primer*. Wiley, Chichester.

IPCC (2021). Climate Change 2021: The Physical Science Basis. Contribution of Working Group I to the Sixth Assessment Report (AR6), Chapter 7 Table 7.15 and Table 7.SM.7: 100-year GWP values for CH_4 (non-fossil) and N_2O. Cambridge University Press.

R packages used by the simulation engine.

Pouillot, R., and Delignette-Muller, M.-L. (2010). *Evaluating variability and uncertainty separately in microbial quantitative risk assessment: a primer using R and a freely-available package, mc2d.* International Journal of Food Microbiology, 142(3), 330–340. (`mc2d` provides `cornode` for the Iman–Conover restricted-pairing reordering used to reproduce the target rank correlation matrix per IPCC Vol.1 Ch.3 §3.2.3.2, and `rtriang` / `rpert` for the triangular and PERT marginals.)

Bates, D., and Maechler, M. (2024). *Matrix*: Sparse and Dense Matrix Classes and Methods. R package. (`Matrix::nearPD`, projection to the nearest positive-definite correlation matrix when a user-supplied $\mathbf{\Sigma}$ is not positive-definite.)

Funded by the Global Methane Hub. Version 1.0.